Set theory for the paper "Extremely projective normed spaces"

A crash course

What this is. The paper "Extremely projective normed spaces" uses, in its sections 6 and 7, a handful of tools from set theory: ordinals and transfinite recursion, the first uncountable ordinal ω_1, closed unbounded and stationary subsets of ω_1, and one theorem of Solovay. These notes explain exactly those tools, assuming only naive set theory and the notion of cardinality. Section 1 introduces well-orders, ordinals and transfinite recursion; section 2 reviews cardinals and the axiom of choice; section 3 treats ω_1 and its closed unbounded sets; section 4 walks through the set-theoretic steps of the paper one by one; and section 5 is a glossary. Exercises come with hints; the ones marked with an asterisk are used in the paper and are worth doing in full. For a textbook treatment see Jech, *Set theory*, chapters 2, 3, 5 and 8, or, for a gentler start, Hrbacek and Jech, *Introduction to set theory*.

1 Well-orders, ordinals and transfinite recursion

1.1 Well-orders

Definition 1.1. A *well-order* is a linearly ordered set $(W, <)$ in which every non-empty subset has a least element.

Example 1.2. $\mathbb{N}$ with its usual order is a well-order. So is any finite linear order. So is $\mathbb{N} \cup \{\infty\}$ with ∞ placed after all natural numbers: a non-empty subset either contains a natural number, and then has a least one, or is $\{\infty\}$. So is "two copies of $\mathbb{N}$ one after the other", the set $\{0,1\} \times \mathbb{N}$ ordered lexicographically with the first coordinate dominant. The sets $\mathbb{Z}$, $\mathbb{Q}$ and $[0,1]$ with their usual orders are not well-orders: $\mathbb{Z}$ has no least element, and $(0,1] \subset [0,1]$ has none.

Two consequences of the definition are used all the time. First, a well-order has no infinite strictly decreasing sequence $x_0 > x_1 > x_2 > \cdots$, since the set $\{x_n : n \in \mathbb{N}\}$ would have no least element. Second, the *principle of induction*: if a property $\Phi(x)$ fails for some $x \in W$, then it fails for a least x, so to prove Φ for all x it suffices to prove $\Phi(x)$ under the assumption that $\Phi(y)$ holds for all $y < x$.

For $x \in W$ the *initial segment* $W_{<x} = \{y \in W : y < x\}$ is again a well-order. The basic structure theorem, which we state without proof, is that well-orders are linearly ordered by "being an initial segment":

Theorem 1.3. *Let V and W be well-orders. Then exactly one of the following holds: V and W are order isomorphic; V is order isomorphic to $W_{<x}$ for a unique $x \in W$; W is order isomorphic to $V_{<y}$ for a unique $y \in V$. In each case the isomorphism is unique.*

1.2 Ordinals

Ordinals are canonical representatives of well-orders, chosen so that the order relation becomes membership. Following von Neumann, put

$$0 = \varnothing, \quad 1 = \{0\}, \quad 2 = \{0, 1\}, \quad \ldots, \quad n = \{0, 1, \ldots, n-1\}, \quad \ldots,$$
$$\omega = \{0, 1, 2, \ldots\} = \mathbb{N}, \quad \omega + 1 = \omega \cup \{\omega\},$$

and so on: each ordinal is the set of all smaller ordinals, ordered by membership. Formally, an *ordinal* is a set α which is well-ordered by $\in$ and *transitive*, meaning that every element of an element of α is an element of α. We write $\beta < \alpha$ for $\beta \in \alpha$. The facts to remember are the following.

(a) Every well-ordered set is order isomorphic to exactly one ordinal, called its *order type*. Ordinals are therefore "the" well-orders, and theorem 1.3 says that any two ordinals α, β satisfy exactly one of $\alpha < \beta$, $\alpha = \beta$, $\beta < \alpha$, where $\alpha < \beta$ also means $\alpha \subsetneq \beta$.

(b) Every element of an ordinal is an ordinal, and the elements of α are exactly the ordinals $\beta < \alpha$. Thus, $\alpha = \{\beta : \beta < \alpha\}$. This is why the paper writes $\ell_1(\delta)$ for the families supported in $\{\beta : \beta < \delta\}$: the ordinal δ *is* that set.

(c) Every set of ordinals has a least element (it is a subset of a well-order), and a supremum, namely its union, which is again an ordinal. There is no set of all ordinals: for every set M of ordinals, $\sup M + 1$ is an ordinal not in M.

(d) The *successor* of α is $\alpha + 1 = \alpha \cup \{\alpha\}$, the least ordinal above α. An ordinal is a *successor ordinal* if it is of the form $\alpha + 1$, and a *limit ordinal* if it is not a successor and not 0. (Some authors count 0 as a limit ordinal; the paper explicitly does not.) A limit ordinal δ is the supremum, that is, the union, of the ordinals below it: $\delta = \bigcup_{\beta<\delta} \beta$. Every ordinal is 0, a successor or a limit.

Example 1.4. The finite ordinals are the natural numbers. The first infinite ordinal is ω, the order type of $\mathbb{N}$, a limit. Then come $\omega + 1, \omega + 2, \ldots$, successors; their supremum $\omega + \omega = \omega \cdot 2$, the order type of two copies of $\mathbb{N}$ one after the other, is the second limit ordinal; then $\omega \cdot 2 + 1, \ldots, \omega \cdot 3, \ldots, \omega \cdot \omega = \omega^2$, the order type of $\mathbb{N} \times \mathbb{N}$ ordered lexicographically; and so on. All of these are countable sets. The limit ordinals among them are exactly the non-zero multiples of ω.

Exercise 1.5. Show that ω and $\omega + 1$ are not order isomorphic. (Hint: one of them has a largest element.) Show that $\omega \cdot 2$ is not order isomorphic to ω. (Hint: in $\omega \cdot 2$ the element ω has infinitely many predecessors but no immediate predecessor.)

Exercise 1.6. Let $(W, <)$ be a linear order with no infinite strictly decreasing sequence. Show that W is a well-order. (Hint: if a non-empty subset M has no least element, choose $x_0 \in M$, then $x_1 < x_0$ in M, and so on; this uses countable choice, see section 2.3.)

1.3 Transfinite induction and recursion

The principle of induction for well-orders becomes, for ordinals, the principle of *transfinite induction*: to prove $\Phi(\alpha)$ for all ordinals α (or for all $\alpha \leq \lambda$) it suffices to prove $\Phi(\alpha)$ assuming $\Phi(\beta)$

for all $\beta < \alpha$. In practice one splits into three cases: $\alpha = 0$; $\alpha = \beta + 1$, where one assumes $\Phi(\beta)$; and α a limit, where one assumes $\Phi(\beta)$ for all $\beta < \alpha$.

More important for the paper is *transfinite recursion*: an object $F(\alpha)$ may be defined for all ordinals $\alpha < \lambda$ by a rule which produces $F(\alpha)$ from the family $(F(\beta))_{\beta<\alpha}$. The theorem of transfinite recursion says that such a rule determines a unique function F on λ. Again one usually gives the rule in three parts: what $F(0)$ is, how $F(\beta+1)$ is obtained from $F(\beta)$, and how $F(\alpha)$ is obtained from $(F(\beta))_{\beta<\alpha}$ at a limit α, typically as a union or a limit.

Example 1.7 (the chains of closed pairs in the paper)**.** In the proof of theorem 6.7 of the paper a family $((A_\alpha, B_\alpha))_{\alpha\le\kappa}$ is defined by transfinite recursion: $(A_0, B_0) = (\varnothing, \varnothing)$; $(A_{\alpha+1}, B_{\alpha+1})$ is obtained from (A_α, B_α) by adding one point and closing off; at a limit α the pair (A_α, B_α) is the union of the previous pairs. In the proof of theorem 6.1 the operators V_δ are defined by recursion on $\delta < \lambda$: the rule producing V_δ uses W_δ and T_δ, which depend only on the $V_{\delta'}$ with $\delta' < \delta$. The accompanying claim $T_\delta(P_\delta) = \ell_1^0(\Gamma_\delta)$ is then proved by transfinite induction on δ, with the limit case handled by the continuity hypothesis.

1.4 Filtrations

The paper calls a family $(\Gamma_\delta)_{\delta\le\lambda}$ of subsets of a set Γ a *filtration* if it is increasing, $\Gamma_0 = \varnothing$, $\Gamma_\lambda = \Gamma$, and it is *continuous*: $\Gamma_\delta = \bigcup_{\delta'<\delta} \Gamma_{\delta'}$ for every limit ordinal $\delta \le \lambda$. Thus, a filtration grows only at successor steps, by the *blocks* $N_\delta = \Gamma_{\delta+1} \setminus \Gamma_\delta$, and at limit steps it merely collects what has been built. Continuity is a genuine condition: the family given by $\Gamma_n = \{0, \dots, n-1\}$ for $n < \omega$, $\Gamma_\omega = \mathbb{N} \cup \{\star\}$ and $\Gamma_{\omega+1} = \mathbb{N} \cup \{\star, \star\star\}$ is increasing but not continuous at ω, because the point $\star$ appears at the limit stage without ever appearing in a block.

Exercise 1.8 (*)**.** Prove by transfinite induction on $\delta \le \lambda$ that for a filtration $\Gamma_\delta = \bigsqcup_{\delta'<\delta} N_{\delta'}$, a disjoint union. (Hint: the successor step is the definition of N_δ; the limit step is continuity.) Deduce that every $x \in \ell_1(\Gamma)$ satisfies $\|x\| = \sum_{\delta'<\lambda} \|x|_{N_{\delta'}}\|$, and that only countably many of the summands are non-zero.

Exercise 1.9 (*)**.** Let $C \subset \omega_1$ be a set and $(c_\alpha)_{\alpha<\mu}$ its increasing enumeration (the elements of C listed in increasing order; see section 2 for why the index set is an ordinal). Show that $\Gamma_\alpha = c_\alpha$ defines a continuous family if and only if C is closed under suprema of its bounded subsets. This is how the paper turns a closed unbounded set into a filtration in the proof of theorem 7.2.

2 Cardinals and the axiom of choice

2.1 Cardinals as initial ordinals

Two sets have the same cardinality if there is a bijection between them. To have a canonical representative of each cardinality one uses ordinals again: the *cardinal* $|X|$ of a set X is the least ordinal which is in bijection with X. For this to make sense X must be in bijection with some ordinal at all, that is, X must admit a well-order; that every set does is the *well-ordering theorem*, equivalent to the axiom of choice, see section 2.3. An ordinal which is the cardinal of some set, that is, which is not in bijection with any smaller ordinal, is called an *initial ordinal*. The finite ordinals are initial, and ω is initial; it is also written $\aleph_0$ when regarded as a cardinal. The ordinals $\omega + 1$, $\omega \cdot 2$, ω^2 of example 1.4 are not initial: they are countable, hence in bijection with ω.

Definition 2.1. ω_1 is the least uncountable ordinal, that is, the least ordinal not in bijection with ω. Regarded as a cardinal it is written $\aleph_1$. Likewise $\omega_2 = \aleph_2$ is the least ordinal of cardinality greater than $\aleph_1$, and so on.

Such ordinals exist: the order types of well-orderings of subsets of $\mathbb{N}$ form a set of ordinals, all countable, and every countable ordinal is among them; the supremum of this set plus one is uncountable, and the least uncountable ordinal exists because every set of ordinals has a least element. By definition, the elements of ω_1 are exactly the countable ordinals:

$$\omega_1 = \{\alpha : \alpha \text{ is a countable ordinal}\}.$$

This double role is the source of most confusion for beginners. The symbol ω_1 denotes both a well-ordered set, the set of countable ordinals in their natural order, and a cardinality $\aleph_1$, the cardinality of that set. When the paper says "let $\Gamma = \omega_1$" it uses the first role: the index set is well-ordered, and the subspaces $\ell_1(\delta)$, $\delta < \omega_1$, are its initial segments. When it says "$|\Gamma| = \aleph_1$" it uses the second role.

The *continuum hypothesis*, the statement $2^{\aleph_0} = \aleph_1$, is independent of the usual axioms and is *not* used anywhere in the paper. Nothing in the paper depends on how large $2^{\aleph_0}$ or $2^{\aleph_1}$ is; the latter only appears as the number of subsets of ω_1.

2.2 Cardinal arithmetic used in the paper

Only the most basic facts are needed, all of which hold for an infinite cardinal κ:

(a) $\kappa \cdot \aleph_0 = \kappa$ and $\kappa + \kappa = \kappa$; more generally $\kappa \cdot \mu = \max(\kappa, \mu)$ for infinite κ, μ;

(b) the set of finite subsets of a set of cardinality κ has cardinality κ, and so has the set of countable subsets of a countable set... which is to say, a countable set has countably many finite subsets, and a set of cardinality κ has $\kappa^{\aleph_0} \geq \kappa$ countable subsets; the paper only needs the finite case and the fact that a countable union of countable sets is countable;

(c) the algebraic dimension of $\ell_1^0(\Gamma)$ is $|\Gamma|$ for infinite Γ, because the unit vectors e_γ form a Hamel basis, and a vector space spanned by κ vectors has dimension at most κ;

(d) the paper writes $\max(\mathrm{dens}(Z), \aleph_0)$ in theorem 3.2 only to cover finite dimensional Z, whose density character is finite.

Exercise 2.2 (*)**.** Show that every element $x \in \ell_1(\Gamma)$ has countable support. (Hint: for each n the set $\{\gamma : |x_\gamma| > 1/n\}$ is finite.) Deduce that every $x \in \ell_1(\omega_1)$ lies in $\ell_1(\delta)$ for some countable ordinal δ, using exercise 3.1 below.

2.3 The axiom of choice

The paper works, as most of functional analysis does, in ZFC: the Zermelo–Fraenkel axioms with the axiom of choice. Choice enters in three forms. The *well-ordering theorem*: every set can be enumerated by an ordinal, that is, written as $\{\gamma_\alpha : \alpha < \kappa\}$ with distinct γ_α; the paper does this with Γ in the proof of theorem 6.7, and with $\kappa = |\Gamma|$ one may take the enumeration to be indexed by the cardinal itself. *Zorn's lemma*: every vector space has a Hamel basis, used throughout. *Countable choice*: a countable union of countable sets is countable, used whenever the paper says that the union of countably many countable supports is countable, and in the proof that a countable set of countable ordinals has a countable supremum. None of these uses is delicate, and the reader may simply take all three as basic facts.

3 The first uncountable ordinal

3.1 Basic properties of ω_1

Lemma 3.1. *Every countable subset of ω_1 is bounded in ω_1: if $M \subset \omega_1$ is countable, then* $\sup M < \omega_1$.

Proof. $\sup M = \bigcup M$ is a countable union of countable sets, hence a countable ordinal. □

This is the single most important property of ω_1. It says that one cannot reach ω_1 in countably many steps: ω_1 has *uncountable cofinality*. Contrast this with every limit ordinal *below* ω_1, which can be reached in countably many steps.

Lemma 3.2 (ladders)**.** *Every limit ordinal $\delta < \omega_1$ is the supremum of a strictly increasing sequence $(\delta_n)_{n\in\mathbb{N}}$ of ordinals below δ.*

Proof. Since δ is countable, enumerate it as $\{\beta_n : n \in \mathbb{N}\}$. Put $\delta_0 = \beta_0$ and $\delta_{n+1} = \max(\delta_n + 1, \beta_{n+1})$; this is below δ because δ is a limit. The sequence is strictly increasing and $\beta_n \le \delta_n$ for all n, so its supremum is δ. □

Such a sequence is called a *ladder* at δ, and a choice of a ladder at every limit ordinal $\delta < \omega_1$, or at every δ in a given set E of limit ordinals, is a *ladder system*. The ladder spaces of section 7 of the paper are built from a ladder system: the vector $v_\delta = \sum_n 2^{-n} e_{\delta_n}$ "climbs" the ladder at δ. Two ladders at different limit ordinals $\alpha < \beta$ have only finitely many rungs in common, because the rungs of the ladder at β eventually exceed α.

Exercise 3.3. Show that for every $\delta < \omega_1$ the subspace $\ell_1(\delta)$ of $\ell_1(\omega_1)$ is separable, and that $\ell_1(\omega_1)$ is not. Show that $\ell_1(\omega_1) = \bigcup_{\delta<\omega_1} \ell_1(\delta)$, while $\ell_1(\delta) \neq \bigcup_{\delta'<\delta} \ell_1(\delta')$ for every limit ordinal $\delta < \omega_1$. (Hint: lemma 3.2.) So the chain $(\ell_1(\delta))_{\delta\le\omega_1}$ is continuous at the top and discontinuous at every limit below; this is the situation the paper's condition (b) of theorem 6.7 rules out for the pieces P_δ.

3.2 Closed unbounded sets

Definition 3.4. A set $C \subset \omega_1$ is *closed* if the supremum of every non-empty countable subset of C belongs to C, equivalently, if the supremum of every strictly increasing sequence in C belongs to C. It is *unbounded* if for every $\beta < \omega_1$ there is $\delta \in C$ with $\delta > \beta$. A *closed unbounded set*, or *club*, is a set which is both.

The word "closed" is the topological one: in the order topology on ω_1 the closed unbounded sets are the closed sets which are unbounded. A club is to be thought of as a *large* set: it contains "almost all" countable ordinals in a sense made precise by the next lemmas. The simplest club is ω_1 itself. The next one is the set Lim of limit ordinals below ω_1: it is closed because a supremum of limit ordinals is a limit ordinal (a supremum of a strictly increasing sequence is never a successor), and unbounded because $\beta + \omega$ is a limit above β.

Lemma 3.5. *Let $h : \omega_1 \to \omega_1$ be any function. The set C_h of ordinals $\delta < \omega_1$ such that $h(\alpha) < \delta$ for all $\alpha < \delta$ is a club. We say that the elements of C_h are* closed under h.

Proof. Closed: let δ be the supremum of a strictly increasing sequence (δ_n) in C_h and $\alpha < \delta$. Then $\alpha < \delta_n$ for some n, so $h(\alpha) < \delta_n \leq \delta$. Unbounded: given β, put $\delta_0 = \beta+1$ and $\delta_{n+1} = \sup\{h(\alpha)+1 : \alpha < \delta_n\} \cup \{\delta_n + 1\}$, a countable ordinal by lemma 3.1. The sequence is strictly increasing; let δ be its supremum, again countable. If $\alpha < \delta$, then $\alpha < \delta_n$ for some n, so $h(\alpha) < \delta_{n+1} < \delta$. Thus, $\delta \in C_h$ and $\delta > \beta$. □

Lemma 3.6. *The intersection of two clubs is a club. Consequently, the intersection of finitely many clubs is a club.*

Proof. Let C, D be clubs. Their intersection is closed, being an intersection of closed sets. To see that it is unbounded, let $\beta < \omega_1$ and choose alternately $c_0 < d_0 < c_1 < d_1 < \cdots$ with $c_n \in C$, $d_n \in D$ and $c_0 > \beta$, which is possible because both sets are unbounded. The two sequences have the same supremum δ, which is countable by lemma 3.1, and $\delta \in C \cap D$ because both sets are closed. □

Lemma 3.7. *Let C be a club and $(c_\alpha)_{\alpha<\omega_1}$ its increasing enumeration. Then $c_\alpha = \sup_{\alpha'<\alpha} c_{\alpha'}$ for every limit ordinal $\alpha < \omega_1$.*

Proof. The enumeration exists because C is a well-ordered set; it is indexed by ω_1 because C is uncountable (unbounded) and every proper initial segment of C is bounded, hence countable. Let α be a limit and $\gamma = \sup_{\alpha'<\alpha} c_{\alpha'}$. Then $\gamma \in C$ because C is closed, $\gamma \geq c_{\alpha'}$ for all $\alpha' < \alpha$, and γ is not among the $c_{\alpha'}$ with $\alpha' < \alpha$, since the sequence has no largest term. Hence γ is the least element of C above all $c_{\alpha'}$, $\alpha' < \alpha$, which is c_α. □

Exercise 3.8 (*)**.** Let $f : \omega_1 \to \omega_1$ be a bijection. Show that $\{\delta < \omega_1 : f(\delta) = \delta\}$, where $f(\delta)$ denotes the image of the set $\delta = \{\beta : \beta < \delta\}$, is a club. (Hint: apply lemma 3.5 to f and to f^{-1} and intersect.) This is the mechanism behind proposition 7.3 of the paper and behind the remark that the enumeration of the index set does not matter.

Exercise 3.9. Show that the intersection of countably many clubs is a club. (Hint: for unboundedness build a sequence which visits every club infinitely often.) The paper needs only finite intersections.

3.3 Stationary sets

Definition 3.10. A set $S \subset \omega_1$ is *stationary* if it intersects every club. A set is *non-stationary* if it is not stationary, that is, if it is disjoint from some club.

If clubs are the large sets, then non-stationary sets are the small ones, and stationary sets are the sets which are not small. Every club is stationary by lemma 3.6; in particular Lim is stationary. Every countable set is non-stationary, being disjoint from the club of ordinals above its supremum. A set containing a club is stationary, but a stationary set need not contain a club: Solovay's theorem below provides stationary sets whose complements are stationary too, and neither of them contains a club.

Lemma 3.11. *A subset of a non-stationary set is non-stationary, and the union of two non-stationary sets is non-stationary. If S is stationary and C is a club, then $S \cap C$ is stationary.*

Proof. The first claim is obvious. If $S \cap C = \varnothing$ and $S' \cap C' = \varnothing$ for clubs C, C', then $(S \cup S') \cap (C \cap C') = \varnothing$ and $C \cap C'$ is a club. For the last claim, if D is a club, then $(S \cap C) \cap D = S \cap (C \cap D) \neq \varnothing$. □

The lemma says that the non-stationary sets form an *ideal*, and therefore the relation "$E \triangle E'$ is non-stationary" is an equivalence relation on subsets of ω_1: this is what "$\Delta(P)$ modulo non-stationary sets" means in section 7 of the paper. Two sets which differ by a non-stationary set are stationary or non-stationary together.

Theorem 3.12 (Solovay; for ω_1 already Ulam)**.** *There is a family $(S_\xi)_{\xi<\omega_1}$ of pairwise disjoint stationary subsets of ω_1 with union ω_1. More generally, every stationary subset of ω_1 is the union of $\aleph_1$ pairwise disjoint stationary sets.*

We do not prove this; it is theorem 8.10 of Jech's book. The paper uses it in the following way. Intersecting each S_ξ with the club Lim we may assume that all S_ξ consist of limit ordinals (lemma 3.11). For $X \subset \omega_1$ put $E_X = \bigcup_{\xi\in X} S_\xi$. If $X \neq Y$, pick $\xi \in X \triangle Y$; then $S_\xi \subset E_X \triangle E_Y$, so $E_X \triangle E_Y$ is stationary, and in particular $E_X \neq E_Y$ modulo non-stationary sets. There are $2^{\aleph_1}$ sets X, hence $2^{\aleph_1}$ pairwise inequivalent sets E_X, and only $E_\varnothing = \varnothing$ is non-stationary.

3.4 Fodor's lemma, for orientation

The paper does not use the following theorem, but it is the first thing one meets after clubs and stationary sets, and it explains why ladder systems on stationary sets are so rigid. A function $g : S \to \omega_1$ on a set S of non-zero ordinals is *regressive* if $g(\alpha) < \alpha$ for all $\alpha \in S$.

Theorem 3.13 (Fodor)**.** *If $S \subset \omega_1$ is stationary and $g : S \to \omega_1$ is regressive, then g is constant on some stationary subset of S.*

Example 3.14. Let $E \subset \mathrm{Lim}$ be stationary and let a ladder L_α at α be chosen for every $\alpha \in E$. Then the ladders cannot be pairwise disjoint: the function $g(\alpha) = \min L_\alpha$ is regressive on E, so by Fodor's lemma it takes the same value γ_0 at two different α, and the corresponding ladders share the rung γ_0. For a non-stationary E pairwise disjoint ladders can exist (exercise). This is the kind of phenomenon behind the fact that the ladder space P_E of the paper is projective exactly when E is non-stationary, although the paper proves that fact without Fodor's lemma.

Exercise 3.15. Let E be a non-stationary set of limit ordinals. Show that there are pairwise disjoint ladders L_α, $\alpha \in E$. (Hint: let C be a club disjoint from E; for $\alpha \in E$ let $c(\alpha)$ be the largest element of C below α, which exists because C is closed and $\alpha \notin C$, and choose the ladder at α inside the interval $(c(\alpha), \alpha)$; distinct α with the same $c(\alpha)$ lie in the same interval $(c(\alpha), c^+)$, where c^+ is the next element of C, and there are only countably many of them, so their ladders can be chosen disjoint by an inductive argument.)

4 Reading guide to sections 6 and 7 of the paper

Theorem 6.1, the transfinite back-and-forth

The data are a filtration $(\Gamma_\delta)_{\delta\le\lambda}$ with countable blocks N_δ and the pieces $P_\delta = P \cap \ell_1(\Gamma_\delta)$. The set-theoretic content is: a transfinite recursion of length λ defining the operators V_δ, and a transfinite induction proving $T_\delta(P_\delta) = \ell_1^0(\Gamma_\delta)$. The successor step is analysis. The limit step is pure bookkeeping, and it is exactly where the continuity condition (ii) is used: at a limit μ the piece P_μ must be the union of the earlier pieces, because the operators T_δ are compatible and the images $\ell_1^0(\Gamma_\delta)$ have union $\ell_1^0(\Gamma_\mu)$. The operator W_δ is a sum over all $\delta' < \delta$, possibly uncountably many; it converges because every $x \in \ell_1(\Gamma_\delta)$ has countable support and meets only countably many blocks (exercise in section 1).

Closed pairs and closing off

Fix the retraction $\sigma : P \to \ell_1^0(S)$, $\pi : \ell_1^0(S) \to P$ and the approximating sequences $p_n^\gamma \to e_\gamma$. A *closed pair* (A, B) is a pair of index sets which is closed under three operations: taking the support of $\pi(e_s)$ for $s \in B$, taking the support of $\sigma(x)$ for $x \in P_A$, and taking the supports of the p_n^γ for $\gamma \in A$. This is the same idea as a subgroup generated by a set, or a Skolem hull in logic: given any sets A_0, B_0, apply the operations, collect the new elements, apply them again, and so on ω times; the union is closed under the operations, because each element of it entered at some finite stage and its images entered at the next stage. The only subtlety, which the paper points out, is the second operation: an element of $P_{A'}$ need not lie in any P_{A_k} (its support may spread over all stages), so closedness under σ is proved by continuity of σ and density of $\bigcup_k P_{A_k}$ rather than by a support argument.

Lemma 6.5 of the paper says that if one starts from a closed pair and adds a countable set, then ω steps of closing off add only countably many points. The reason is lemma 6.4: over a closed pair, a countable extension of A changes P_A by a quotient of countable dimension, so only countably many new supports are needed at each stage. This is the analogue of the downward Löwenheim–Skolem theorem: a countable set generates a countable closed structure.

Theorem 6.7, the chain of closed pairs

Enumerate $\Gamma = \{\gamma_\alpha : \alpha < \kappa\}$ (well-ordering theorem) and define closed pairs by transfinite recursion: add γ_α and close off at successor steps, take unions at limits. Lemma 6.6 says that a union of closed pairs along a chain is closed, and this is what makes the limit step work and gives the continuity condition for free. The resulting family $(A_\alpha)_{\alpha \le \kappa}$ is a filtration with countable blocks, and the three hypotheses of theorem 6.1 are verified by the three lemmas on closed pairs. Nothing about the size of Γ matters here; this is why the paper no longer needs an induction on cardinals.

Theorem 7.2, the criterion for ω_1

Here $\Gamma = \omega_1$ and $P_\delta = P \cap \ell_1(\delta)$. The theorem says: P is extremely projective if and only if the pieces have countable dimension and the chain (P_δ) is continuous on a club, that is, the set $\Delta(P)$ of limit ordinals at which it jumps is non-stationary.

For the forward direction one runs the chain of closed pairs with $\gamma_\alpha = \alpha$. Each A_α is a countable subset of ω_1 (countable increments, countable unions of countable sets). Define $f(\alpha) = \sup A_\alpha + 1$ and $g(\beta) =$ the first α with $\beta \in A_\alpha$. On the club of limit ordinals closed under f and g (lemmas 3.5 and 3.6) one has $A_\delta = \delta$: closure under f gives $A_\delta \subset \delta$, closure under g gives $\delta \subset A_\delta$. At such δ the union lemma turns P_δ into a union of earlier pieces, so $\delta \notin \Delta(P)$.

For the converse one needs a filtration. Let

$$h(\gamma) = \sup \bigcup\nolimits_n \operatorname{supp}(p_n^\gamma) + 1.$$

The club C_1 of ordinals closed under h consists of ordinals at which the piece is dense; a club C_2 misses $\Delta(P)$; the enumeration of $(C_1 \cap C_2) \cup \{0\}$ is continuous (lemma 3.7) and defines a filtration of ω_1 with countable blocks. Continuity of the chain of pieces along it holds because its limit points are limit ordinals in C_2, and the quotients are countable because the pieces are. Theorem 6.7 finishes the proof.

Proposition 7.3, invariance

Given a bounded bijection T of $\ell_1(\omega_1)$, the function $F(\delta) = \sup_{\beta<\delta}(\sup \operatorname{supp} T(e_\beta) + 1)$ records how far T moves the coordinates below δ. On the club of limit ordinals closed under F and under the analogous G for T^{-1}, T maps $\ell_1(\delta)$ onto itself, hence maps the chain of pieces of P onto that of $T(P)$, and jumps correspond. So $\Delta(P)$ and $\Delta(T(P))$ agree on a club, that is, modulo a non-stationary set. Applied to a permutation of the index set this shows that the choice of enumeration is immaterial.

Theorem 7.4, ladder spaces

For a set E of limit ordinals the ladder space is $P_E = \ell_1^0(\omega_1) + \operatorname{span}\{v_\alpha : \alpha \in E\}$ with v_α climbing a ladder at α (lemma 3.2). The computation $\Delta(P_E) = E$ uses only that two ladders share finitely many rungs. Then theorem 7.2 says that P_E is extremely projective exactly when E is non-stationary, proposition 7.3 says that $P_E \ncong P_{E'}$ when $E \triangle E'$ is stationary, and Solovay's theorem 3.12 produces $2^{\aleph_1}$ sets E_X with pairwise stationary symmetric differences.

Remark 7.5, almost free abelian groups

An abelian group is free if it has a basis; it is $\aleph_1$-free if all its countable subgroups are free. Pontryagin's criterion and Hill's lemma say that a group of cardinality $\aleph_1$ is free if and only if it is the union of a continuous ω_1-chain of countable free subgroups with free successive quotients; Eklof's invariant is the set of limit stages at which such a chain cannot be continued with a free quotient, taken modulo non-stationary sets; and ladder systems on stationary sets give $\aleph_1$-free groups which are not free. The paper's theorem 6.7, definition 7.1 and theorem 7.4 are the exact counterparts, with "countable dimension" in place of "free". The reader who wants to see where this leads above $\aleph_1$ may consult chapters IV and VII of Eklof and Mekler.

5 Glossary

well-order
a linear order in which every non-empty subset has a least element (section 1).

ordinal
a canonical well-order; the set of all smaller ordinals (section 1).

successor ordinal, limit ordinal
$\alpha+1$, respectively a non-zero ordinal which is not a successor; a limit is the union of the ordinals below it (section 1).

ω
the first infinite ordinal, the set of natural numbers; as a cardinal, $\aleph_0$.

ω_1
the first uncountable ordinal, the set of countable ordinals; as a cardinal, $\aleph_1$ (section 2).

ω_2, $\aleph_2$
the next initial ordinal and cardinal; mentioned in the paper only in the open questions.

transfinite induction, recursion
proving a property, respectively defining an object, at α from the ordinals below α (section 1).

enumeration by an ordinal
a bijection $\alpha \mapsto \gamma_\alpha$ from an ordinal κ onto a set Γ; exists by the well-ordering theorem (section 2.3).

filtration, block
an increasing continuous family of subsets of Γ starting at $\varnothing$ and ending at Γ; the differences of consecutive members (section 1).

cardinality of Γ, $|\Gamma|$
the least ordinal in bijection with Γ (section 2).

density character
the least cardinality of a dense subset; for $\ell_1(\Gamma)$ with Γ infinite it is $|\Gamma|$.

closed unbounded set, club
a subset of ω_1 closed under countable suprema and unbounded (section 3).

closed under h
an ordinal δ with $h(\alpha) < \delta$ for all $\alpha < \delta$; these ordinals form a club (lemma 3.5).

stationary, non-stationary
meeting every club, respectively disjoint from some club (section 3).

modulo non-stationary sets
the equivalence relation "$E \triangle E'$ is non-stationary" (lemma 3.11).

ladder, ladder system
a strictly increasing ω-sequence converging to a limit ordinal; a choice of one at every limit ordinal of a given set (lemma 3.2).

Solovay's theorem
ω_1 splits into $\aleph_1$ disjoint stationary sets (theorem 3.12).

Fodor's lemma
a regressive function on a stationary set is constant on a stationary set; not used in the paper (section 3).

continuum hypothesis
$2^{\aleph_0} = \aleph_1$; not used in the paper.

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